Differential R-to-V Converter

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Reference: M. Ahmad *et al.*, "An Auto-Calibrated Resistive Measurement System With Low Noise Instrumentation ASIC," *IEEE Journal of Solid-State Circuits*, vol. 55, no. 11, pp. 3036–3050, Nov. 2020.

Motivation

- Sensors such as **temperature sensors** and **strain gauges** change resistance with actuation.
- Most flexible wearable devices commonly use **resistive sensors**.
- Most processing circuits and ADCs work with **voltage signals**, hence a need for **R-to-V conversion**.
- A **differential R-to-V converter** minimizes mismatch effects.

Half-Bride over Wheatstone

- Using **Half-Bridge** instead of Wheatstone bridge for measurement:
  - Improved accuracy due to lesser mismatch effect.
  - Wider measurement range — Not limited by supply voltage.
  - Lower power consumption.

(Proposed solution : Auto-calibration enabled by programmable current source and INA gain)

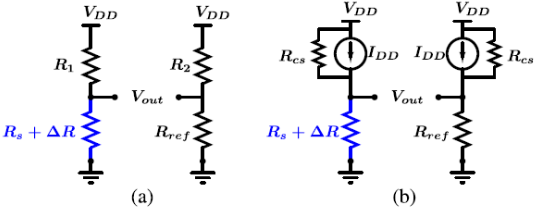


Figure: (a)Wheatstone bridge (b)Current-excited Half-Bridge

Comparison of Wheatstone Bridge and Current Excitation Bridge

Parameters	**Wheatstone bridge**	**Current excitation bridge**
Output voltage (V_{out})	$V_{DD} \cdot \frac{k\frac{\Delta R}{R_s}}{(1+k)\cdot\left(1+k+\frac{\Delta R}{R_s}\right)}$	$I_{DD} \cdot \frac{\Delta R}{R_s} \cdot R_s$
Sensitivity $\left(\frac{\partial V_{out}}{\partial(\Delta R/R_s)}\right)$	$V_{DD} \cdot \frac{k}{(1+k+\frac{\Delta R}{R_s})^2}$	$I_{DD} \cdot R_s$
PSRR $\left(\frac{\partial V_{DD}}{\partial V_{out}}\right)$	$\frac{(1+k)\left(1+k+\frac{\Delta R}{R_s}\right)}{k\frac{\Delta R}{R_s}}$	$\frac{k}{\frac{\Delta R}{R_s}}$
k	$\frac{R_1}{R_s} = \frac{R_2}{R_{ref}} = 1$	$\frac{R_{cs}}{R_s} > 1000$
Max Sensitivity at$(\Delta R/R_s = 0)$	$V_{DD}/4$	$I_{DD} \cdot R_s = V_{DD}/2$
PSRR at $(\Delta R/R_s = 100\%)$	15 dB	60 dB

Why Square Wave Current Excitation?

- **Current excitation** increases dynamic range.
- DC excitation suffers from thermoelectric effects.
- Generating pure sinusoidal excitation is power-hungry and requires complex circuitry.
- Square wave excitation:
  - Simpler to generate.
  - Contains harmonics handled by Differential Ratiometric Operation (DRO).

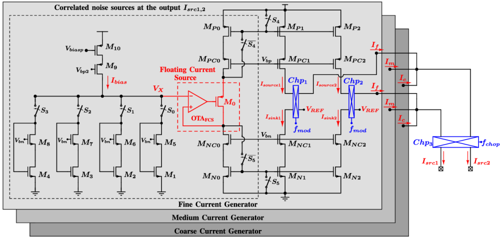


Fig. Square wave current generation

Circuit Overview

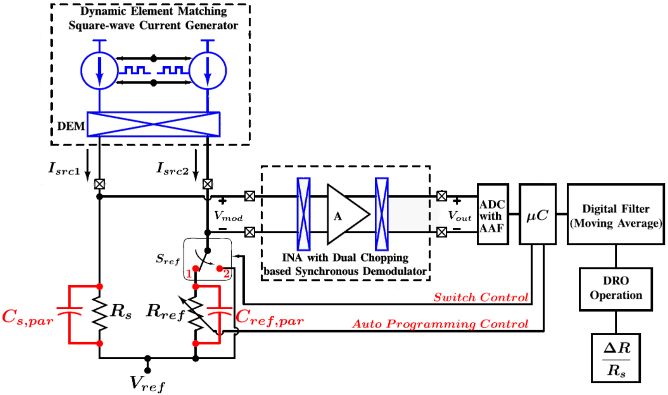


Figure: Proposed differential R-to-V converter system for $\Delta R/R_s$ measurement.

Circuit Overview

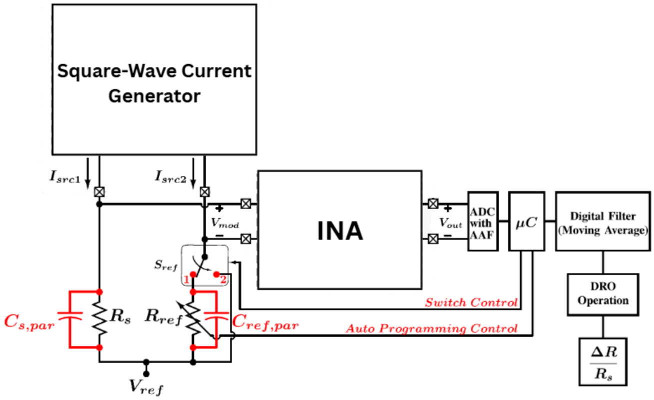


Figure: Proposed differential R-to-V converter system for $\Delta R/R_s$ measurement.

Instrumentational Amplifier (INA)

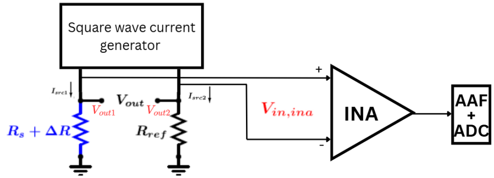


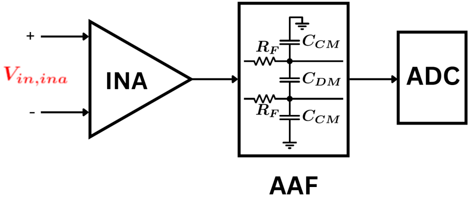
Fig: INA

$$
\begin{aligned}
I_{\text{src1}} &= I_{\text{src2}} \\
R_s &= R_{\text{ref}} \\
V_{\text{out1}} &= I_s \left(R_s + \Delta R_s\right) \\
V_{\text{out2}} &= I_s R_s \\
V_{\text{out}} &= V_{\text{out1}} - V_{\text{out2}} = I_s \Delta R_s
\end{aligned}
$$

- Provides high Common Mode Rejection Ratio (CMRR).

Anti-Aliasing Filter

- Square waves contain odd harmonics of the fundamental frequency.
- Without filtering, harmonics alias into the signal band, leading to distortion.
- The AAF ensures accurate ADC sampling and signal reconstruction.



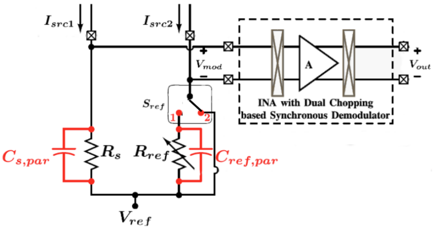
Filter response showing attenuation of higher harmonics.

Differential Ratiometric Operation (DRO)

Phase 1: Maximizing sensitivity

$$V_{out} = I_s \frac{\Delta R_s * R_s}{R_s}$$

$$S = I_s * R_s$$

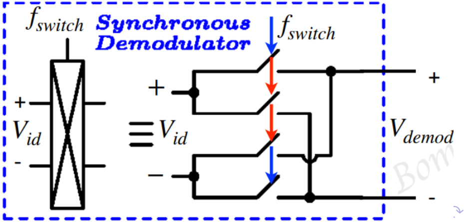


Phase 2: Baseline resistance measurement

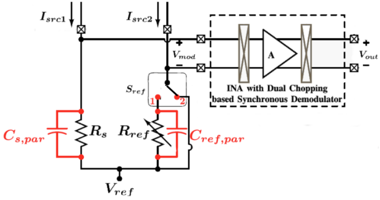
$$V_{\text{out}|\text{phase 2}} = I_{\text{src}}(t) \times R_s \times A \times V_{\text{mod}}(t)$$

$$I_{\text{src}}(t) = \frac{4I}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \sin\left((2n-1)\omega_0 t\right)$$

$$V_{\text{mod}}(t) = \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \sin\left((2n-1)\omega_0 t\right)$$



Differential Ratiometric Operation (DRO)



Phase 2: Baseline resistance measurement

$$V_{\text{out}|\text{phase 2}} = \left(\frac{4}{\pi}\right)^2 IR_sA \sum_{n=1}^{\infty} \frac{\sin^2((2n-1)\omega_0 t)}{(2n-1)^2}$$

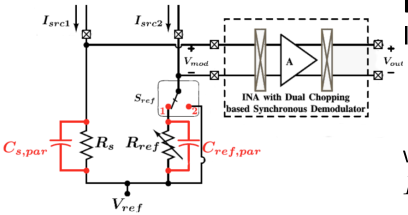
$$= \left(\frac{4}{\pi}\right)^2 IR_sA \cdot \frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{1}{(2n-1)^2} - \frac{2\cos(2(2n-1)\omega_0 t)}{(2n-1)^2} \right)$$

Oscillatory part is removed by Moving filter

$$V_{\text{out}|\text{phase 2}} = \left(\frac{4}{\pi}\right)^2 I \times A \times R_s \times D$$

$$\text{where } D = \sum_{n=1}^{\infty} \frac{1}{2} \cdot \frac{1}{(2n-1)^2}$$

Differential Ratiometric Operation (DRO)



Phase 3: Mismatch calibration

In phase 3, ideally V_{out} should be zero, as R_{ref} is adjusted to R_s

$$V_{\text{out}|\text{ phase } 3} = \left(\tfrac{4}{\pi}\right)^2 A[I\delta R + \delta I(R_s - \delta R)]D$$

where; $R_s - R_{ref} = \delta R$

$I_{src1} - I_{src2} = \delta I$

Phase 4: Final measurement

$$V_{\text{out}|\text{ phase } 4} = \left(\frac{4}{\pi}\right)^2 A[I(\Delta R + \delta R) + \delta I(R_s - \delta R)]D$$

Final Output:

$$\frac{\Delta R_s}{R_s} = \frac{V_{\text{out}|\text{ phase } 4} - V_{\text{out}|\text{ phase } 3}}{V_{\text{out}|\text{ phase } 2}}$$

Effect of Parasitic Capacitance

$$V_{\text{out}} = \left(\frac{4}{\pi}\right)^2 AI \sum_{n=1}^{\infty} \frac{1}{2(2n-1)^2} (Z_s - Z_{\text{ref}})$$

Sensor branch: $R_s \parallel C_{s,\text{par}}$
Reference branch:$R_{\text{ref}} \parallel C_{ref}$, par

$Z_{eq}(j\omega) = \frac{R}{1+j\omega C}$

$Z_s(n) = \frac{R_s}{1+j(2n-1)\omega R_s C_{s,\text{par}}}$
$Z_{ref}(n) = \frac{R_{\text{ref}}}{1+j(2n-1)\omega R_{\text{ref}} C_{\text{ref},\text{par}}}$

$\omega_{\text{clk}} \ll \frac{1}{R_s C_{s,\text{par}}}, \quad \omega_{\text{clk}} \ll \frac{1}{R_{\text{ref}} C_{\text{ref,par}}}$

$$V_{\text{out}} \approx \left(\frac{4}{\pi}\right)^2 \frac{A \cdot I}{2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} (R_s - R_{\text{ref}})$$

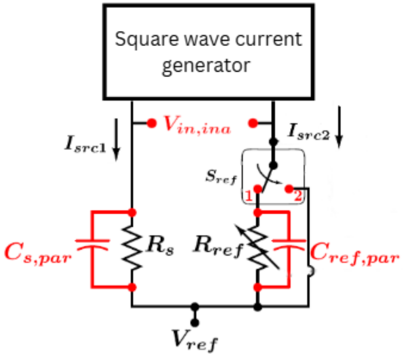
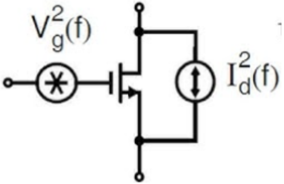
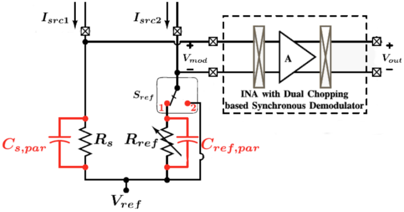


Fig.Effect of parasitic capacitance

Frequency Tradeoff

- Reducing excitation frequency decreases parasitic effects.
- However, lower frequencies increase MOSFET flicker noise.
- $V_g^2(f) = \dfrac{K}{WLC_{ox}f}$
- Thermal Noise: $I_d^2(f) = 4kT(\frac{2}{3})g_m$





Specifications

Parameter	Value
Excitation Signal	AC Current (Square)
Technology Node	180nm
Supply	1.8V
Base Resistor R_s	1kΩ – 500kΩ
Excitation Current (Prog.)	$0.1\mu A$ – $44.4\mu A$
CMRR of INA	$>$110 dB
Power	33.3 μW @ 100 nA 357 μW @ 44.4 μA
ADC resolution	15 bit

Max $\Delta R/R$	Min $\Delta R/R$
$\pm 135\%$ @ $R_s = 10$ kΩ, $I = 4.4\,\mu$A	26 ppm @ $R_s = 10$ kΩ, $I = 4.4\,\mu$A
$\pm 120\%$ @ $R_s = 0.5$ MΩ, $I = 100\,\text{nA}$	72 ppm @ $R_s = 0.5$ MΩ, $I = 100\,\text{nA}$

Any Questions?

Comments

- Add required figure on every slide.
- Draw switch positions in figure.

According to the feedback received in presentation- 1, the slides are modified.